

REMARKS ON QUASI-EINSTEIN MANIFOLDS AND APPLICATIONS

ALCIDES DE CARVALHO AND W. O. COSTA-FILHO

ABSTRACT. In this article, we revisit previous results on m -quasi-Einstein manifolds. First, we complete a statement of [4] by characterizing the structure when m and λ have opposite signs, and we generalize Corollary 1.5 of Bahuaud et al. [1] to arbitrary values of m . As an application, we provide a classification of closed six-dimensional locally conformally flat m -quasi-Einstein manifolds admitting nontrivial divergence-free vector fields.

1. INTRODUCTION

An n -dimensional Riemannian manifold (M^n, g) , with $n \geq 2$, is called an m -quasi-Einstein manifold for a nonzero constant $m \in \mathbb{R}$ if there exist a smooth vector field X on M and a constant $\lambda \in \mathbb{R}$ such that the following equation holds:

$$\text{Ric} + \frac{1}{2}\mathcal{L}_X g - \frac{1}{m}X^\flat \otimes X^\flat = \lambda g, \quad (1)$$

where Ric denotes the Ricci tensor, $\mathcal{L}_X g$ is the Lie derivative of the metric g in the direction of X , $X^\flat$ is the 1-form metrically dual to X , and $\otimes$ denotes the tensor product. An m -quasi-Einstein manifold is called trivial when X is identically zero.

Special cases of m -quasi-Einstein metrics encompass several important geometric structures, including Einstein metrics, static metrics, gradient Ricci solitons, and, more generally, the base manifolds of Einstein warped products for various positive integer values of m , see [5–7, 14]. When m is negative, the framework also includes conformally Einstein metrics and conformally Einstein products [3]. Owing to the significance of these distinct but interrelated metrics, and the fact that they can be unified through variation of the parameter m , the quasi-Einstein condition can be naturally interpreted as interpolating between these different geometric structures. Notably, in the limiting case $m = \infty$ when M is closed (i.e., compact without boundary), play a central role in Perelman’s formulation of the Ricci flow [16]. In this case, equation (1) reduces to

$$\text{Ric} + \frac{1}{2}\mathcal{L}_X g = \lambda g.$$

This is precisely the defining equation of a Ricci soliton, which arises naturally as a self-similar solution to the Ricci flow and plays a key role in understanding the behavior of the Ricci flow, especially in the context of singularities.

It turns out that many standard results of Riemannian geometry have extensions to the setting of m -quasi-Einstein manifolds. In particular, the Myers compactness theorem was extended in the context of the so-called m -Bakry–Émery Ricci curvature in many ways [13, 20]. On the other hand, general relativity has spurred a comprehensive study of these metrics. For instance, in black hole spacetimes the intrinsic geometry of an extremal horizon is described by a Riemannian metric and a vector field on an n -dimensional cross section of the horizon that satisfies a quasi-Einstein structure. See [15] for more details and references in this direction.

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While many approaches focus on the gradient case—when the vector field X is the gradient of a smooth function—non-gradient m -quasi-Einstein metrics have gained increasign attention due to their relevance in near-horizon geometry and in the modeling of perfect fluid spacetimes. In this framework, the work of Bahuaud et al. [1] provides illustrative examples. See also [2, 3, 8, 21] and the references therein.

Recall that a smooth vector field X on a Riemannian manifold (M^n, g) is called a Killing vector field if $\mathcal{L}_X g = 0$. Such fields have profound topological implications. In particular, the assumption that a compact manifold admits a Killing vector field has been extensively explored in the literature from various perspectives. When this additional symmetry is present, it imposes strong constraints on the structure of an m -quasi-Einstein manifold in different geometric and analitical contexts, as we now discuss.

In [1], the authors studied compact solutions to equation (1) and showed that if X is incompressible and $m \neq -2$, then X is a Killing vector field (for the exceptional value $m = -2$, see [10, 12]). In addition to that, very recently E. Cochran [10] and W. O. Costa-Filho [12] independently proved that in a closed m -quasi-Einstein manifold, the scalar curvature S of g is constant if and only if X is a Killing vector field.

It is natural to ask whether there is some rigidity and classification results for compact m -quasi-Einstein manifolds, when $dX^\flat$ is closed. If $\lambda = 0$, Chruściel–Reall–Tod [9] proved that X must vanish identically and (M^n, g) is Ricci flat. For $\lambda > 0$, Bahuaud et al. [2] show that X must necessarily be gradient. In the $\lambda < 0$ case, Wylie [21] shows that $X = 0$ or (M^n, g) is isometric to the product of a circle and an Einstein manifold.

Our first contribution in this work is observed an incomplete statement in Barros–Ribeiro [4], namely item (3) of Lemma 2, by showing that the conclusion holds only under the additional assumption that m and λ have the same sign. Moreover, when m and λ do not share the same sign, we show that the manifold splits as $\mathbb{S}^1 \times N^{n-1}$, with N being Einstein.

Secondly, we extend certain corollary of Bahuaud et al. [1] by removing restrictions on m and obtaining sharper splitting results under simple sign conditions, see corollary 1.8 in [1].

Finally, we classify six-dimensional closed locally conformally flat m -quasi-Einstein manifolds admitting nontrivial divergence-free vector fields, determining the exact scalar curvature values compatible with such structures and identifying precisely when the manifold splits as $\mathbb{S}^1 \times N^{n-1}$ with N Einstein.

2. PRELIMINARIES

In this section, we present some auxiliary results concerning m -quasi-Einstein metrics. Throughout, all Riemannian manifolds (M^n, g) are assumed to be connected and oriented. We begin by recalling the following basic fact.

Lemma 1. *Let (M^n, g) be a Riemannian manifold, and let X be a smooth vector field. If X is parallel, i.e., $\nabla X = 0$, then the associated 1-form $X^\flat$ is closed, i.e., $dX^\flat = 0$. Conversely, if X is a Killing vector field and $dX^\flat = 0$, then X is parallel.*

Proof. Using the Cartan formula for the exterior derivative of a 1-form, together with the symmetry and metric-compatibility of the Levi-Civita connection, we have

$$dX^\flat(Y, Z) = \langle \nabla_Y X, Z \rangle - \langle \nabla_Z X, Y \rangle \quad \text{for all } Y, Z \in \mathfrak{X}(M). \quad (2)$$

If X is parallel, then clearly $dX^\flat = 0$.

Conversely, if X is Killing, its covariant derivative is skew-symmetric:

$$\langle \nabla_Y X, Z \rangle + \langle \nabla_Z X, Y \rangle = 0 \quad \text{for all } Y, Z \in \mathfrak{X}(M).$$

Then, from (2),

$$dX^\flat(Y, Z) = \langle \nabla_Y X, Z \rangle - \langle \nabla_Z X, Y \rangle = 2\langle \nabla_Y X, Z \rangle.$$

Hence, if $dX^\flat = 0$, it follows that $\nabla X = 0$, so X is parallel. $\square$

On a Riemannian manifold (M^n, g) and for any smooth vector field X , the tensors $\mathcal{L}_X g$ and $dX^\flat$ correspond to the symmetric and skew-symmetric parts of ∇X . This leads to a concise norm identity.

Proposition 2. *For every $X \in \mathfrak{X}(M)$,*

$$4|\nabla X|^2 = |\mathcal{L}_X g|^2 + |dX^\flat|^2. \quad (3)$$

Proof. From the definitions,

$$(\mathcal{L}_X g)(Y, Z) = \langle \nabla_Y X, Z \rangle + \langle \nabla_Z X, Y \rangle, \quad dX^\flat(Y, Z) = \langle \nabla_Y X, Z \rangle - \langle \nabla_Z X, Y \rangle,$$

we immediately get

$$(\mathcal{L}_X g)(Y, Z) + dX^\flat(Y, Z) = 2\langle \nabla_Y X, Z \rangle.$$

Summing over a basis, this yields

$$|\mathcal{L}_X g|^2 + |dX^\flat|^2 = 4|\nabla X|^2,$$

which proves the desired result. $\square$

We will need the following relation due to Barros and Ribeiro [4, page 215] (see also Lemma 2.3 of [1]). Here, Δ denotes the Laplace–Beltrami operator on M^n .

Lemma 3. *Let (M^n, g, X, λ) be an m -quasi-Einstein manifold. Then we have,*

$$\frac{1}{2}\Delta|X|^2 = |\nabla X|^2 + \frac{2}{m}|X|^2 \operatorname{div} X - \operatorname{Ric}(X, X).$$

We close this section by quoting the following expression, as observed by Bahuaud et al. [1]. For sake of completeness, we will give here a proof of it.

Lemma 4. *Let (M^n, g, X, λ) be a closed m -quasi-Einstein manifold, with X divergence-free. Then,*

$$|dX^\flat|^2 = 4|X|^2 \left(\lambda + \frac{|X|^2}{m} \right). \quad (4)$$

Proof. Since X is incompressible, it follows from Theorem 1.1 of [12] that X is Killing, with $|X|$ constant. Therefore, from equality (3) together with Lemma 3 and equation (1), it follows that

$$|dX^\flat|^2 = 4|\nabla X|^2 = 4\operatorname{Ric}(X, X) = 4|X|^2 \left(\lambda + \frac{|X|^2}{m} \right).$$

$\square$

Note that the constants $|X|^2$ and $|dX^\flat|^2$ depend only on the scalar curvature S of M^n , the parameter m , and the constant λ . Indeed, taking the trace of equation (1) and using the fact that the scalar curvature S is constant (as established in [12]), we obtain

$$|X|^2 = m(S - n\lambda).$$

Substituting this equality into (4) yields an identity for $|dX^\flat|^2$ in terms of S , λ , and m .

3. RELATED RESULTS

We begin this section by revisiting item (3) of Lemma 2 in [4]. We observed that the original statement omits an assumption—namely, that m and λ must have the same sign—without which the conclusion does not hold. We next deal with the remaining case where m and λ have opposite signs.

The refined statement is given below.

Proposition 5. *Let (M^n, g, X, λ) be an m -quasi-Einstein manifold with $\nabla X = 0$. Then one of the following holds:*

- (1) *If m and λ have the same sign, then $X = 0$.*
- (2) *If m and λ have opposite signs, assume that M^n is closed. Then, either $X = 0$ or (M^n, g) is isometric to the Riemannian product*

$$(\mathbb{S}^1 \times N^{n-1}, d\theta^2 + g_N),$$

where (N^{n-1}, g_N) is an Einstein manifold with

$$\text{Ric}_{g_N} = \lambda g_N,$$

and the vector field X is given by

$$X = \pm \sqrt{-m\lambda} \partial_\theta.$$

Proof. Initially, taking into account that $\nabla X = 0$ we have $|X|$ is constant and $\text{div } X = 0$.

Hence, the identity in Lemma 3 gives $\text{Ric}(X, X) = 0$. Now, using equation (1), we obtain

$$\frac{1}{m}|X|^4 + \lambda|X|^2 = 0. \quad (5)$$

If $m\lambda \geq 0$, equation (5) forces $X = 0$, concluding the proof for this case. At this point, in [4] the authors assume $X \neq 0$ and show that $R(X, Y) = 0$ for all vector fields Y . However, this condition alone does *not* imply that the manifold is Ricci-flat.

Assume $X \neq 0$. From equation (5), we obtain

$$\lambda = -\frac{1}{m}|X|^2 \neq 0.$$

Using the fact that $\text{div } X = 0$ and substituting this value of λ into the expression for the norm of $dX^\flat$ given in equation (4), we conclude that $dX^\flat = 0$, i.e., the 1-form $X^\flat$ is closed. Thus the result follows directly from Theorem 1.1 in [21]. □

Corollary 6. *Let (M^n, g, X, λ) be an m -quasi-Einstein manifold. Assume that X is a divergence-free vector field and that $dX^\flat = 0$. Then X is parallel, and one of the following holds:*

- (1) *If m and λ have the same sign, then $X = 0$.*
- (2) *If m and λ have opposite signs, assume that M^n is closed. Then, either $X = 0$ or (M^n, g) is isometric to the Riemannian product $(\mathbb{S}^1 \times N^{n-1}, d\theta^2 + g_N)$, where (N^{n-1}, g_N) is Einstein with $\text{Ric}_{g_N} = \lambda g_N$, and*

$$X = \pm \sqrt{-m\lambda} \partial_\theta.$$

Proof. Since X is divergence-free vector field, it follows from Theorem 1.1 of [12] that X is Killing. On the other hand, since X is Killing and $dX^\flat = 0$, Lemma 1 implies that X is parallel. Then Proposition 5 applies, which yields the two possibilities depending on whether m and λ have the same or opposite signs. □

We note that Theorem 1 in [4] asserts that the manifold (M^n, g) is Einstein under therein stated assumptions. However, this conclusion holds only when m and λ have the same sign; otherwise, the proof actually establishes only the parallelism of the vector field X . Accordingly, the result can be more accurately stated as follows:

Theorem 7. *Let (M^n, g, X) , with $n \geq 3$, be a closed m -quasi-Einstein manifold. Assume that one of the following conditions holds:*

- (1) $\int_M \text{Ric}(X, X) dM \leq \frac{2}{m} \int_M |X|^2 \text{div } X dM$.
- (2) X is a conformal vector field and $\int_M \text{Ric}(X, X) dM \leq 0$.
- (3) $|X|$ is constant and $\int_M \text{Ric}(X, X) dM \leq 0$.

Therefore, the vector field X is parallel in all cases; moreover, it vanishes if m and λ have the same sign, while it may be nonzero otherwise.

Theorem 7 establishes the parallelism of X , which in turn allows us to apply Proposition 5 and deduce the following classification depending on the signs of m and λ , leading either to an Einstein structure or to a Riemannian product splitting $(\mathbb{S}^1 \times N^{n-1}, d\theta^2 + g_N)$, where (N^{n-1}, g_N) is Einstein. To be precise:

Corollary 8. *Let (M^n, g, X, λ) be an m -quasi-Einstein manifold satisfying the assumptions of Theorem 7. Then one of the following holds:*

- (1) *If m and λ have the same sign, then $X = 0$ and (M^n, g) is an Einstein manifold.*
- (2) *If m and λ have opposite signs, assume that M^n is closed. Then, either $X = 0$ or (M^n, g) is isometric to the Riemannian product*

$$(\mathbb{S}^1 \times N^{n-1}, d\theta^2 + g_N),$$

where (N^{n-1}, g_N) is Einstein with $\text{Ric}_{g_N} = \lambda g_N$, and

$$X = \pm \sqrt{-m\lambda} \partial_\theta.$$

Proof. Since, in all cases, X is parallel by Theorem 7, the conclusion follows directly from Corollary 6. □

As a consequence of the result above, we refine Corollary 1.8 in Bahuaud et al. [1] by removing any assumption that the parameter m lies within a specific interval.

We prove that the same conclusion holds for arbitrary values of m , provided m and λ have opposite signs. Specifically:

Corollary 9. *Let (M^n, g, X, λ) be a closed m -quasi-Einstein manifold satisfying $|X|^2 = -m\lambda$. Then, X is parallel. In particular, (M, g) splits isometrically as*

$$(\mathbb{S}^1 \times N^{n-1}, d\theta^2 + g_N),$$

where (N^{n-1}, g_N) is an Einstein manifold with $\text{Ric}_{g_N} = \lambda g_N$, and

$$X = \pm \sqrt{-m\lambda} \partial_\theta.$$

Proof. Since the norm of X is constant, it follows from Theorem 7 that if

$$\int_M \text{Ric}(X, X) dM \leq 0,$$

then X must be parallel. On the other hand, the condition $|X|^2 = -m\lambda$ implies that m and λ have opposite signs. Hence, by item (2) of Corollary 8, the manifold M splits isometrically as claimed.

We now claim that

$$\int_M \text{Ric}(X, X) dM = 0,$$

or more specifically, $\text{Ric}(X, X) = 0$. Indeed, applying both sides of (1) to the pair (X, X) , we obtain:

$$\text{Ric}(X, X) + \frac{1}{2}(\mathcal{L}_X g)(X, X) - \frac{1}{m}|X|^4 = \lambda|X|^2.$$

Since the norm of X is constant, we have:

$$(\mathcal{L}_X g)(X, X) = \langle \nabla_X X, X \rangle = 0.$$

Thus, the identity simplifies to:

$$\text{Ric}(X, X) = \lambda|X|^2 + \frac{1}{m}|X|^4.$$

Now, using the assumption $|X|^2 = -m\lambda$, we substitute:

$$\text{Ric}(X, X) = \lambda(-m\lambda) + \frac{1}{m}(-m\lambda)^2 = -m\lambda^2 + m\lambda^2 = 0.$$

This completes the proof of the claim. □

Next, we conclude this section by recalling the classical *Yano identity* (see, for instance, [17]), which states that for any smooth vector field Y on a closed Riemannian manifold M^n ,

$$\frac{1}{2} \int_M |\mathcal{L}_Y g|^2 dM = \int_M (|\nabla Y|^2 + (\text{div } Y)^2 - \text{Ric}(Y, Y)) dM. \quad (6)$$

4. QUASI-EINSTEIN STRUCTURES WITH CLOSED $X^\flat$

As discussed by Colling and Dunajski in [11], the case of quasi-Einstein structures with closed one-form $X^\flat$ and negative parameter $m < 0$ exhibits substantially different geometric properties from the classical setting $m > 0$.

Quasi-Einstein structures (M, g, X, λ) on a closed manifold M with $m > 0$ and closed but non-exact $X^\flat$ have been completely classified by Bahuaud *et al.* [2] and Wylie [21]. This condition is of particular significance when $m = 2$, since the corresponding near-horizon geometries are static and possess a local AdS_3 symmetry [15].

Theorem 10. [2, 21] *Let (M^n, g, X, λ) be a closed quasi-Einstein manifold with $m > 0$ and $X^\flat$ closed but not exact. Then $\lambda < 0$, and (M, g, X, λ) is a quotient of $(\widetilde{M}, \tilde{g}, \tilde{X})$ by a discrete group of isometries, where*

$$\widetilde{M} = N \times \mathbb{R}, \quad \tilde{g} = g_N \oplus dt^2, \quad \tilde{X} = \sqrt{-m\lambda} \frac{\partial}{\partial t}.$$

Here, t denotes the coordinate on $\mathbb{R}$, and (N, g_N) is a simply connected negative Einstein manifold satisfying $\text{Ric}(g_N) = \lambda g_N$. By a quotient of a quasi-Einstein manifold we mean a solution satisfying $\tilde{X} = \pi^* X$ and $\tilde{g} = \pi^* g$, where $\pi : \widetilde{M} \rightarrow M$ is the quotient map.

In contrast, the solutions corresponding to $m < 0$ turn out to be significantly less rigid. In particular, the vector field X does not need to be parallel, as demonstrated by the explicit examples presented in Section 3.2 of [11]. By applying their Theorem 1.1, Colling and Dunajski established that for $m \leq 2 - n$, the general solution is not a direct product of $\mathbb{R}$ with an Einstein manifold, but rather a *warped product* of $\mathbb{R}$ with a (gradient) quasi-Einstein manifold.

Theorem 11 ([11, Theorem 2.5]). *Let (M^n, g, X, λ) be a closed quasi-Einstein manifold of dimension n with $m \leq 2 - n$ and $X^\flat$ closed but not exact. Then $\lambda > 0$, and (M, g, X, λ) is a quotient of $(\widetilde{M}, \tilde{g}, \tilde{X})$ by a discrete group of isometries, where*

$$\widetilde{M} = N \times \mathbb{R}, \quad \tilde{g} = g_N \oplus e^{\frac{-2}{m}u} dt^2, \quad \tilde{X}^\flat = dt + du.$$

Here, t denotes the coordinate on $\mathbb{R}$, u is a smooth function on N , and (N, g_N) is a simply connected quasi-Einstein manifold with vector field $X_N^\flat = \frac{m+1}{m} du$, and parameters $(m+1, \lambda)$.

The objective of the next result is to identify a range of parameters for which the vector field X is necessarily parallel.

Theorem 12. *Let (M^n, g, X, λ) be a closed m -quasi-Einstein manifold with $m \neq 2$, and suppose that $X^\flat$ is closed but not exact. Set $s := |X|^2$ and consider the quadratic polynomial*

$$F(s) := \frac{m-2}{m} s^2 + 2\lambda(m-1)s + m^2 \lambda^2.$$

Its discriminant is $\Delta = 4\lambda^2 > 0$, so F has two distinct real roots

$$t_1 = -\lambda m, \quad t_2 = -\frac{\lambda m^2}{m-2},$$

and write $\alpha := \min\{t_1, t_2\}$, $\beta := \max\{t_1, t_2\}$.

Assume the following pointwise condition on M :

- *If $m > 2$ or $m < 0$, then $s(x) \in [\alpha, \beta]$ for all $x \in M$;*
- *If $0 < m < 2$, then $s(x) \notin (\alpha, \beta)$ for all $x \in M$.*

Then X is parallel, and $|X|^2 = -m\lambda$. Consequently (M, g) splits isometrically as

$$(\mathbb{S}^1 \times N^{n-1}, d\theta^2 + g_N),$$

where (N^{n-1}, g_N) is Einstein with $\text{Ric}_{g_N} = \lambda g_N$, and

$$X = \pm \sqrt{-m\lambda} \partial_\theta.$$

Proof. From the argument of Lemma 3.2 in [2] (valid by the same computations for any real m), since $X^\flat$ is closed but not exact, we have

$$\text{div } X - |X|^2 = m\lambda \quad \text{on } M.$$

Integrating over M and using the divergence theorem yields

$$-\int_M |X|^2 dM = m\lambda \text{Vol}(M),$$

hence

$$m\lambda = -\frac{1}{\text{Vol}(M)} \int_M |X|^2 dM.$$

Because $X^\flat$ is not exact, we have $X \not\equiv 0$, so $\int_M |X|^2 dM > 0$. Thus $m\lambda < 0$ and $\lambda \neq 0$. Next, using identity (3) together with Lemma 3 and Yano's identity, we get

$$2 \int_M |\nabla X|^2 dM = \int_M \left[(\text{div } X)^2 - \frac{2}{m} |X|^2 \text{div } X \right] dM.$$

Substituting $\text{div } X = |X|^2 + m\lambda$, we obtain the following

$$\begin{aligned}
2 \int_M |\nabla X|^2 dM &= \int_M \left[(\operatorname{div} X)^2 - \frac{2}{m} |X|^2 \operatorname{div} X \right] dM \\
&= \int_M \left[(|X|^2 + m\lambda)^2 - \frac{2}{m} |X|^2 (|X|^2 + m\lambda) \right] dM \\
&= \int_M \left[\frac{m-2}{m} |X|^4 + 2\lambda(m-1) |X|^2 + m^2 \lambda^2 \right] dM \\
&= \int_M F(s) dM.
\end{aligned}$$

By the pointwise hypothesis we have $F(s) \leq 0$ on M , hence the right-hand side is ≤ 0 , while the left-hand side is ≥ 0 . Therefore, both sides vanish, so $|\nabla X|^2 \equiv 0$ and X is parallel.

Since X is parallel and $X \neq 0$, X has no zeros and $|X|^2$ is constant; substituting $\operatorname{div} X = |X|^2 + m\lambda$ shows $|X|^2 = -m\lambda > 0$. The final part of the theorem is an immediate consequence of Corollary 9. This completes the proof. $\square$

Remark 13. The result for $m > 0$ and $\lambda < 0$ is due to Wylie [21] and does not require the assumption on $|X|$. We include it here for completeness, since our argument also covers this case.

5. LOCALLY CONFORMALLY FLAT m -QUASI-EINSTEIN MANIFOLDS WITH NONTRIVIAL DIVERGENCE-FREE VECTOR FIELD

We now extend the study of locally conformally flat m -quasi-Einstein manifolds to six dimensions. In particular, we focus on those manifolds admitting a nontrivial divergence-free vector field X . Note that a Riemannian manifold (M^n, g) of dimension $n \geq 4$ is locally conformally flat if and only if its Weyl tensor vanishes.

In Bahuaud et al. [1], a classification was obtained for 4-dimensional closed locally conformally flat m -quasi-Einstein manifolds with divergence-free vector field X that is not identically vanishing. Their results show that:

- (i) If $m\lambda > 0$, then $|X|^2 \geq 2m\lambda$, and (M, g) is conformally flat if and only if $|X|^2 = 2m\lambda$;
- (ii) If $\lambda = 0$, then (M, g) is not locally conformally flat;
- (iii) If $m\lambda < 0$, then $|X|^2 \geq -m\lambda$, and (M, g) is locally conformally flat if and only if $|X|^2 = -m\lambda$, in which case it splits as $\Sigma \times S^1$, with Σ a compact hyperbolic surface.

Very recently, R. Sharma [19] presented a more refined classification (including case $m = -2$) also in four dimensional case.

In the present work, we extend this type of classification to dimension six. More precisely, we classify closed six-dimensional locally conformally flat m -quasi-Einstein manifolds under the assumption that associated vector field X is divergence-free and non-trivial.

In a closed six-dimensional Riemannian manifold (M^6, g) , assuming that the scalar curvature S is constant and (M^6, g) is locally conformally flat, the Euler characteristic $\chi(M)$ can be expressed in terms of geometry using the Gauss–Bonnet–Chern formula. This formula, due to Raulot [18], reads:

$$384\pi^3 \chi(M) = \frac{2}{75} S^3 \operatorname{Vol}(M, g) - \frac{4}{5} S \int_M |\operatorname{Ric}^\circ|^2 dM, \quad (7)$$

where Ric° denotes the traceless part of the Ricci tensor.

Theorem 14. *Let (M^6, g, X, λ) be a closed, 6-dimensional, m -quasi-Einstein manifold such that X is divergence-free and not identically zero. If (M^6, g) is locally conformally flat, then*

the scalar curvature S is either identically zero or constant and equal to one of the following values:

(i) $S = 5\lambda$, or

(ii) $S = \frac{15}{2}\lambda$.

In case (i), the manifold (M, g) splits isometrically as

$$(M, g) \cong (\mathbb{S}^1 \times N^5, d\theta^2 + g_N),$$

where (N^5, g_N) is an Einstein manifold with

$$\text{Ric}_{g_N} = \lambda g_N,$$

and

$$X = \pm \sqrt{-m\lambda} \partial_\theta.$$

Proof. Let (M^6, g, X, λ) be a closed, 6-dimensional, m -quasi-Einstein manifold such that the vector field X is divergence-free and not identically zero. As we already know from [1] (or also [10, 12] for other cases), this condition implies that X is a Killing vector field. Moreover, the scalar curvature S must be constant and consequently the norm $|X|$ is also constant.

Then, equation (1) simplifies to

$$\text{Ric} = \lambda g + \frac{1}{m} X^\flat \otimes X^\flat. \quad (8)$$

Taking the squared norm of both sides and using the orthogonal decomposition of the Ricci tensor, we obtain

$$|\mathring{\text{Ric}}|^2 = |\text{Ric}|^2 - \frac{1}{6} S^2 = 6\lambda^2 + \frac{2\lambda}{m} |X|^2 + \frac{1}{m^2} |X|^4 - \frac{1}{6} S^2.$$

It follows from the trace in equation (8) that

$$S = 6\lambda + \frac{1}{m} |X|^2. \quad (9)$$

Thus, we conclude that

$$|\text{Ric}|^2 = \frac{1}{6} S^2 + \frac{5}{6m^2} |X|^4,$$

and therefore

$$|\mathring{\text{Ric}}|^2 = \frac{5}{6m^2} |X|^4, \quad (10)$$

which shows that $|\mathring{\text{Ric}}|^2$ is constant.

At this point observed that, since X is not identically zero and has constant norm, it must be nowhere vanishing. Therefore, by the Poincaré–Hopf theorem, the Euler characteristic satisfies $\chi(M) = 0$. Thus, the Gauss–Bonnet–Chern formula (7) becomes

$$\frac{2}{75} S^3 \text{Vol}(M, g) - \frac{4}{5} S |\mathring{\text{Ric}}|^2 \text{Vol}(M, g) = 0.$$

Since $\text{Vol}(M, g) > 0$, we conclude that either $S = 0$ or

$$\frac{2}{75} S^2 = \frac{4}{5} |\mathring{\text{Ric}}|^2, \quad \text{that is,} \quad |\mathring{\text{Ric}}|^2 = \frac{1}{30} S^2.$$

Combining this with equation (10), we obtain

$$S^2 = \frac{25}{m^2} |X|^4.$$

Hence, the scalar curvature satisfies

$$S = \pm \frac{5}{m} |X|^2.$$

Combining this with the traced equation (9), we find that

$$|X|^2 = -m\lambda \quad \text{or} \quad |X|^2 = \frac{3}{2}m\lambda,$$

which implies

$$S = 5\lambda \quad \text{or} \quad S = \frac{15}{2}\lambda.$$

In the first case, the manifold M splits isometrically by Corollary 9, as we wanted to show. □

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UNIVERSIDADE FEDERAL DE PERNAMBUCO, DEPARTAMENTO DE MATEMATICA, AV. JORN. ANÍBAL FERNANDES, S/N, CIDADE UNIVERSITÁRIA, RECIFE - PE, BRAZIL, CEP: 50740-560.

Email address: alcides.junior@ufpe.br

UNIVERSIDADE FEDERAL DE ALAGOAS, CAMPUS ARAPIRACA, AV. MANOEL SEVERINO BARBOSA, S/N, BOM SUCESSO, ARAPIRACA - AL, BRAZIL, CEP: 57309-005.

Email address: wagner.filho@arapiraca.ufal.br